

Employee Attrition Prediction

Employee attrition is one of the biggest challenges organizations face because it affects productivity, increases recruitment and training costs, and impacts overall employee morale. The purpose of this project was to understand why employees leave the organization and identify practical ways to improve employee retention. Rather than relying on assumptions, the project analyzed historical employee data to uncover patterns and provide evidence-based recommendations that HR teams can use to make better workforce decisions.

The analysis began by examining employee information across different departments, job roles, salary levels, work-life balance ratings, years of experience, and length of service within the company. This helped identify the areas where employee turnover was most severe and the factors that contributed the most to employees leaving the organization.

One of the first findings was that employee attrition varies significantly across departments. The **Sales department recorded the highest attrition rate (20.63%)**, followed by **Human Resources (19.05%)**, while **Research & Development had the lowest attrition rate (13.84%)**. This indicates that retention challenges are concentrated within specific business units rather than being an organization-wide issue.

Looking deeper into individual job roles, the analysis found that **Sales Representatives experienced the highest attrition rate (39.76%)**, meaning nearly four out of every ten employees in this role eventually left the company. **Laboratory Technicians (23.94%)** and **Human Resources employees (23.08%)** also showed relatively high turnover. These findings suggest that workload, performance expectations, career growth opportunities, or job-related pressures may differ significantly between roles.

Compensation was another important factor influencing employee retention. Employees who remained with the organization earned an average monthly income of approximately **\$6,833**, whereas employees who left earned an average of **\$4,787**. Although salary alone does not determine whether an employee stays or leaves, the results suggest that competitive compensation plays an important role in improving long-term employee commitment.

The project also examined the relationship between work-life balance and attrition. Employees who rated their work-life balance as **"Bad"** experienced an attrition rate of **31.25%**, which was more than twice the rate of employees who reported a healthier work-life balance. This finding highlights the importance of employee well-being and suggests that reducing excessive workloads and improving work-life balance could significantly reduce employee turnover.

Another clear pattern emerged when employee tenure was analyzed. The largest number of resignations occurred during the **first year of employment (59 employees)**, followed by

employees with **two years of service (27 employees)**. This indicates that the first year is the most critical period for employee retention and emphasizes the importance of effective onboarding, mentoring, and early employee engagement.

To support HR decision-making, predictive models were developed to estimate the likelihood of employee attrition based on historical data. Three different approaches were evaluated and compared.

Among the models tested:

- **Gradient Boosting** achieved the highest prediction accuracy (**86.4%**).
- **Random Forest** achieved an accuracy of **85.7%**.
- **Logistic Regression** achieved an accuracy of **73.8%** while providing the greatest transparency in explaining why employees leave.

Additional performance evaluation showed that each model had different strengths. Logistic Regression achieved the highest **Recall (0.67)**, making it better at identifying employees who may leave. Random Forest achieved the highest **Precision (0.64)**, meaning its predictions were more reliable with fewer false alarms. Gradient Boosting achieved the strongest overall discrimination ability with a **ROC-AUC score of 0.82**, making it the best-performing model overall.

Although Gradient Boosting delivered the best prediction performance, Logistic Regression remained highly valuable because it clearly explained which employee characteristics had the greatest influence on attrition. This level of transparency is particularly important for HR professionals because it allows them to understand the reasons behind employee turnover instead of simply receiving a prediction.

The feature importance analysis revealed the factors that most strongly influence employee attrition. **OverTime** was identified as the single most important risk factor, indicating that employees working excessive overtime are much more likely to leave the company. **Monthly Income** was the second strongest factor, showing that higher salaries improve employee retention. Other important factors included **Total Working Years, Years at Company, Job Level, Marital Status, Age, Distance From Home, Stock Option Level, and Job Satisfaction**. Together, these findings demonstrate that employee retention is influenced by a combination of workload, compensation, career progression, employee satisfaction, and personal circumstances.

Based on these insights, several practical business recommendations were developed. The organization should prioritize reducing excessive overtime, improving employee work-life balance, strengthening onboarding programs for new employees, providing mentoring during the first year, reviewing compensation for high-turnover roles, and creating targeted retention strategies for departments such as Sales and Human Resources. Regular employee feedback

sessions, career development opportunities, workload assessments, and stay interviews can further help identify employee concerns before they lead to resignation.

It is also important to recognize the limitations of predictive analysis. While the models successfully identify employees who may be at higher risk of leaving, they cannot measure personal factors such as relationships with managers, workplace culture, family responsibilities, career aspirations, or external job opportunities. Therefore, these predictions should be used to support HR decision-making rather than replace human judgment.

Overall Conclusion

This project demonstrates how employee data can be transformed into meaningful business insights that help organizations improve workforce planning and employee retention. The findings consistently show that **overtime, lower salaries, poor work-life balance, early career stage, and specific job roles particularly Sales Representatives are the strongest contributors to employee attrition.**

By addressing these issues through improved onboarding, competitive compensation, healthier workloads, stronger employee engagement, and department-specific retention initiatives, organizations can reduce turnover, improve employee satisfaction, lower hiring costs, and build a more stable and productive workforce. Rather than simply predicting which employees may leave, this project provides HR leaders with actionable insights that can help create a better workplace and improve long-term organizational success.